1) a)

$$\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^{7}=\cos\frac{7\pi}{6}+\mathrm{i}\sin\frac{7\pi}{6}=\cos\left(-\frac{5\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{5\pi}{6}\right)$$

1) b)

$$\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)^{5}=\cos\frac{10\pi}{3}+\mathrm{i}\sin\frac{10\pi}{3}=\cos\left(-\frac{2\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{3}\right)$$

1) c)

$$\left[2\left(\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\right)\right]^{11}=2^{11}\left(\cos\frac{11\pi}{5}+\mathrm{i}\sin\frac{11\pi}{5}\right)=2048\left(\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\right)$$

1) d)

$$\left[3\left(\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}\right)\right]^{8}=3^{8}\left(\cos\frac{16\pi}{5}+\mathrm{i}\sin\frac{16\pi}{5}\right)=6561\left(\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$$

1) e)

$$\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{-7}=\cos\left(-\frac{7\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{7\pi}{4}\right)=\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}$$

1) f)

$$\left[2\left(\cos\frac{4\pi}{9}+\mathrm{i}\sin\frac{4\pi}{9}\right)\right]^{-4}=2^{-4}\left(\cos\left(-\frac{16\pi}{9}\right)+\mathrm{i}\sin\left(-\frac{16\pi}{9}\right)\right)$$
$$=\frac{1}{16}\left(\cos\frac{2\pi}{9}+\mathrm{i}\sin\frac{2\pi}{9}\right)$$

2) a)

$$\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{15}=\cos\frac{15\pi}{4}+\mathrm{i}\sin\frac{15\pi}{4}=\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}\mathrm{i}$$

2) b)

$$\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)^{10}=\cos\frac{20\pi}{3}+\mathrm{i}\sin\frac{20\pi}{3}=-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$$

2) c)

$$\left[4\left(\cos\frac{3\pi}{8}+\mathrm{i}\sin\frac{3\pi}{8}\right)\right]^{6}=4^{6}\left(\cos\frac{18\pi}{8}+\mathrm{i}\sin\frac{18\pi}{8}\right)=4096\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}\right)$$
$$=2048\sqrt{2}+2048\sqrt{2}\mathrm{i}$$

2) d)

$$\left[\mathrm{i}\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)\right]^5=\mathrm{i}^5\left(\cos\frac{25\pi}{4}+\mathrm{i}\sin\frac{25\pi}{4}\right)=\mathrm{i}\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}\right)$$
$$=-\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\mathrm{i}$$

2) e)

$$\left[-3\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\right]^6=(-3)^6\left(\cos\frac{6\pi}{3}+\mathrm{i}\sin\frac{6\pi}{3}\right)=729$$

2) f)

$$\left[2\left(\cos\frac{5\pi}{6}+\mathrm{i}\sin\frac{5\pi}{6}\right)\right]^{-4}=2^{-4}\left(\cos\left(-\frac{10\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{10\pi}{3}\right)\right)=-\frac{1}{32}+\frac{\sqrt{3}}{32}\mathrm{i}$$

3) a)

$$\left(\cos\frac{\pi}{5}+\mathrm{i}\sin\frac{\pi}{5}\right)^7=\cos\frac{7\pi}{5}+\mathrm{i}\sin\frac{7\pi}{5}=\cos\left(-\frac{3\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{3\pi}{5}\right)=\mathrm{e}^{-\frac{3\pi}{5}\mathrm{i}}$$

3) b)

$$\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^{10}=\cos\frac{10\pi}{3}+\mathrm{i}\sin\frac{10\pi}{3}=\cos\left(-\frac{2\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{3}\right)=\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$$

3) c)

$$\left[3\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)\right]^5=3^5\left(\cos\frac{5\pi}{2}+\mathrm{i}\sin\frac{5\pi}{2}\right)=243\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)=243\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$$

3) d)

$$\left[3\left(\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}\right)\right]^8=3^8\left(\cos\frac{16\pi}{5}+\mathrm{i}\sin\frac{16\pi}{5}\right)=6561\left(\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)$$
$$=6561\mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$$

3) e)

$$\left[2\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)\right]^{-4}=2^{-4}\left(\cos\left(-\frac{4\pi}{6}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{6}\right)\right)$$
$$=\frac{1}{16}\left(\cos\left(-\frac{2\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)=\frac{1}{16}\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$$

3) f)

$\left[-3\left(\cos\frac{5\pi}{8}+\mathrm{i}\sin\frac{5\pi}{8}\right)\right]^{-3}=(-3)^{-3}\left(\cos\left(-\frac{15\pi}{8}\right)+\mathrm{i}\sin\left(-\frac{15\pi}{8}\right)\right)$

$=-\frac{1}{27}\left(\cos\frac{\pi}{8}+\mathrm{i}\sin\frac{\pi}{8}\right)=\frac{1}{27}\left(\cos\left(-\frac{7\pi}{8}\right)+\mathrm{i}\sin\left(-\frac{7\pi}{8}\right)\right)$

$=\frac{1}{27}\mathrm{e}^{-\frac{7\pi}{8}\mathrm{i}}$

4) a)

Let $z=1+\sqrt{3}\mathrm{i}$ so that $|z|=\sqrt{1^2+3}=2$ and $\arg z=\arctan\left(\frac{\sqrt{3}}{1}\right)=\frac{\pi}{3}$

$z=2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$

So $\left(1+\sqrt{3}\mathrm{i}\right)^7=\left[2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\right]^7=2^7\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right)$

$=128\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$

4) b)

Let $z=1-\mathrm{i}$ so that $|z|=\sqrt{1^2+(-1)^2}=\sqrt{2}$ and $\arg z=-\arctan\left(\frac{1}{1}\right)=-\frac{\pi}{4}$

$z=\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

So $(1-\mathrm{i})^{10}=\left[\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)\right]^{10}=\left(\sqrt{2}\right)^{10}\left(\cos\left(-\frac{10\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{10\pi}{4}\right)\right)$

$=32\left(\cos\left(-\frac{\pi}{2}\right)+\mathrm{i}\sin\left(-\frac{\pi}{2}\right)\right)$

4) c)

Let $z=2\mathrm{i}$ so that $|z|=2$ and $\arg z=\frac{\pi}{2}$

$z=2\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)$

So $(2\mathrm{i})^{20}=\left[2\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)\right]^{20}=2^{20}\left(\cos\frac{20\pi}{2}+\mathrm{i}\sin\frac{20\pi}{2}\right)$

$=1048576(\cos 0+\mathrm{i}\sin 0)$

4) d)

Let $z = -8 + 8\sqrt{3}\text{i}$ so that $|z| = \sqrt{(-8)^2 + \left(8\sqrt{3}\right)^2} = 16$

and $\arg z = \pi - \arctan\left(\frac{8\sqrt{3}}{8}\right) = \frac{2\pi}{3}$

$z = 16\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)$

So $\left(-8 + 8\sqrt{3}\text{i}\right)^5 = \left[16\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)\right]^5 = 16^5\left(\cos\frac{10\pi}{3} + \text{i}\sin\frac{10\pi}{3}\right)$

$= 1048576\left(\cos\frac{10\pi}{3} + \text{i}\sin\frac{10\pi}{3}\right)$

$= 1048576\left(\cos\left(-\frac{2\pi}{3}\right) + \text{i}\sin\left(-\frac{2\pi}{3}\right)\right)$

5)

Let $z = -1 + \sqrt{3}\text{i}$ so that $|z| = \sqrt{(-1)^2 + 3} = 2$ and $\arg z = \pi - \arctan\left(\frac{\sqrt{3}}{1}\right) = \frac{2\pi}{3}$

$z = 2\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)$

So $\left(-1 + \sqrt{3}\text{i}\right)^{-4} = \left[2\left(\cos\frac{2\pi}{3} + \text{i}\sin\frac{2\pi}{3}\right)\right]^{-4} = 2^{-4}\left(\cos\left(-\frac{8\pi}{3}\right) + \text{i}\sin\left(-\frac{8\pi}{3}\right)\right)$

$= -\frac{1}{32} - \frac{\sqrt{3}}{32}\text{i}$

6)

Let $z = 2 - 2\text{i}$ so that $|z| = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$ and $\arg z = -\arctan\left(\frac{2}{2}\right) = -\frac{\pi}{4}$

$z = 2\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \text{i}\sin\left(-\frac{\pi}{4}\right)\right)$

So $(2 - 2\text{i})^{-5} = \left[2\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \text{i}\sin\left(-\frac{\pi}{4}\right)\right)\right]^{-5} = \left(2\sqrt{2}\right)^{-5}\left(\cos\frac{5\pi}{4} + \text{i}\sin\frac{5\pi}{4}\right)$

$= -\frac{1}{256} - \frac{1}{256}\text{i}$

7) a)

$\cos\theta - \text{i}\sin\theta = \cos(-\theta) + \text{i}\sin(-\theta)$

7) b)

$(\cos(-\theta) + \text{i}\sin(-\theta))^8 = \cos(-8\theta) + \text{i}\sin(-8\theta)$

8)

$\cos\frac{\pi}{5} - \mathrm{i}\sin\frac{\pi}{5} = \cos\left(-\frac{\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{\pi}{5}\right)$

So $\left(\cos\left(-\frac{\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{\pi}{5}\right)\right)^9 = \cos\left(-\frac{9\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{9\pi}{5}\right) = \cos\frac{\pi}{5} + \mathrm{i}\sin\frac{\pi}{5}$

9) a)

Let $z = 1 + \mathrm{i}$ so that $|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\arg z = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$

$z = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

So $(1+\mathrm{i})^8 = \left[\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\right]^8 = \left(\sqrt{2}\right)^8\left(\cos\frac{8\pi}{4} + \mathrm{i}\sin\frac{8\pi}{4}\right) = 16$

Let $z = 1 - \mathrm{i}$ so that $|z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$ and $\arg z = -\arctan\left(\frac{1}{1}\right) = -\frac{\pi}{4}$

$z = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

So $(1-\mathrm{i})^8 = \left[\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)\right]^8 = \left(\sqrt{2}\right)^8\left(\cos\left(-\frac{8\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{8\pi}{4}\right)\right) = 16$

$\therefore (1+\mathrm{i})^8 + (1-\mathrm{i})^8 = 16 + 16 = 32$

9) b)

Let $z = \sqrt{3} + \mathrm{i}$ so that $|z| = \sqrt{3 + 1^2} = 2$ and $\arg z = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$z = 2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$

So $\left(\sqrt{3}+\mathrm{i}\right)^7 = \left[2\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)\right]^7 = 2^7\left(\cos\frac{7\pi}{6} + \mathrm{i}\sin\frac{7\pi}{6}\right) = -64\sqrt{3} - 64\mathrm{i}$

Let $z = \sqrt{3} - \mathrm{i}$ so that $|z| = \sqrt{3 + (-1)^2} = 2$ and $\arg z = -\arctan\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$

$z = 2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)$

So $\left(\sqrt{3}-\mathrm{i}\right)^7 = \left[2\left(\cos\left(-\frac{\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{\pi}{6}\right)\right)\right]^7 = 2^7\left(\cos\left(-\frac{7\pi}{6}\right) + \mathrm{i}\sin\left(-\frac{7\pi}{6}\right)\right) =$

$-64\sqrt{3} + 64\mathrm{i}$

$\therefore \left(\sqrt{3}+\mathrm{i}\right)^7 + \left(\sqrt{3}-\mathrm{i}\right)^7 = -64\sqrt{3} - 64\mathrm{i} + -64\sqrt{3} + 64\mathrm{i} = -128\sqrt{3}$

10)

$(\cos 9\theta + \mathrm{i}\sin 9\theta)(\cos 6\theta + \mathrm{i}\sin 6\theta) = (\cos\theta + \mathrm{i}\sin\theta)^9(\cos\theta + \mathrm{i}\sin\theta)^6$

$= (\cos\theta + \mathrm{i}\sin\theta)^{15}$

$= \cos 15\theta + \mathrm{i}\sin 15\theta$

11) a)

$$\left(\cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3}\right)\left(\cos\frac{7\pi}{3}+\mathrm{i}\sin\frac{7\pi}{3}\right)=\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^4\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^7$$
$$=\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)^{11}$$
$$=\cos\frac{11\pi}{3}+\mathrm{i}\sin\frac{11\pi}{3}$$
$$=\cos\left(-\frac{\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{\pi}{3}\right)$$

11) b)

$$\left(\cos\frac{17\pi}{4}+\mathrm{i}\sin\frac{17\pi}{4}\right)\left(\cos\frac{5\pi}{4}+\mathrm{i}\sin\frac{5\pi}{4}\right)=\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{17}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^5$$
$$=\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)^{22}$$
$$=\cos\frac{22\pi}{4}+\mathrm{i}\sin\frac{22\pi}{4}$$
$$=\cos\left(-\frac{\pi}{2}\right)+\mathrm{i}\sin\left(-\frac{\pi}{2}\right)$$

11) c)

$$\left(\cos\frac{7\pi}{2}+\mathrm{i}\sin\frac{7\pi}{2}\right)\left(\cos\frac{13\pi}{2}-\mathrm{i}\sin\frac{13\pi}{2}\right)=\left(\cos\frac{7\pi}{2}+\mathrm{i}\sin\frac{7\pi}{2}\right)\left(\cos\left(-\frac{13\pi}{2}\right)-\mathrm{i}\sin\left(-\frac{13\pi}{2}\right)\right)$$
$$=\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)^7\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)^{-13}$$
$$=\left(\cos\frac{\pi}{2}+\mathrm{i}\sin\frac{\pi}{2}\right)^{-6}$$
$$=\cos\left(-\frac{6\pi}{2}\right)+\mathrm{i}\sin\left(-\frac{6\pi}{2}\right)$$
$$=\cos\pi+\mathrm{i}\sin\pi$$

12) a)

$$\left(\cos\frac{\pi}{6}+\mathrm{i}\sin\frac{\pi}{6}\right)^4\left(\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}\right)^5$$
$$=\left(\cos\frac{4\pi}{6}+\mathrm{i}\sin\frac{4\pi}{6}\right)\left(\cos\left(-\frac{\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)^5$$
$$=\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)\left(\cos\left(-\frac{5\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{5\pi}{3}\right)\right)$$
$$=\cos(-\pi)+\mathrm{i}\sin(-\pi)$$
$$\therefore z=\cos\pi+\mathrm{i}\sin\pi$$

12) b)

$$z=-1$$

13) a)

$$\frac{(\cos 9\theta + \mathrm{i}\sin 9\theta)^4}{(\cos 4\theta + \mathrm{i}\sin 4\theta)^6} = \frac{\cos 36\theta + \mathrm{i}\sin 36\theta}{\cos 24\theta + \mathrm{i}\sin 24\theta} = \cos(36\theta - 24\theta) + \mathrm{i}\sin(36\theta - 24\theta)$$
$$= \cos 12\theta + \mathrm{i}\sin 12\theta$$

13) b)

$$\frac{(\cos 5\theta + \mathrm{i}\sin 5\theta)^8}{[2(\cos 3\theta + \mathrm{i}\sin 3\theta)]^6} = \frac{\cos 40\theta + \mathrm{i}\sin 40\theta}{64(\cos 18\theta + \mathrm{i}\sin 18\theta)}$$
$$= \frac{1}{64}(\cos(40\theta - 18\theta) + \mathrm{i}\sin(40\theta - 18\theta))$$
$$= \frac{1}{64}(\cos 22\theta + \mathrm{i}\sin 22\theta)$$

14) a)

$$\frac{\cos\frac{4\pi}{11} + \mathrm{i}\sin\frac{4\pi}{11}}{\cos\frac{5\pi}{11} - \mathrm{i}\sin\frac{5\pi}{11}} = \frac{\cos\frac{4\pi}{11} + \mathrm{i}\sin\frac{4\pi}{11}}{\cos\left(-\frac{5\pi}{11}\right) + \mathrm{i}\sin\left(-\frac{5\pi}{11}\right)} = \cos\left(\frac{4\pi}{11} - -\frac{5\pi}{11}\right) + \mathrm{i}\sin\left(\frac{4\pi}{11} - -\frac{5\pi}{11}\right)$$
$$= \cos\frac{9\pi}{11} + \mathrm{i}\sin\frac{9\pi}{11}$$

14) b)

$$\frac{\left(\cos\frac{9\pi}{14} + \mathrm{i}\sin\frac{9\pi}{14}\right)^6}{\left(\cos\frac{3\pi}{14} - \mathrm{i}\sin\frac{3\pi}{14}\right)^4} = \frac{\left(\cos\frac{9\pi}{14} + \mathrm{i}\sin\frac{9\pi}{14}\right)^6}{\left(\cos\left(-\frac{3\pi}{14}\right) + \mathrm{i}\left(-\frac{3\pi}{14}\right)\right)^4} = \frac{\cos\frac{54\pi}{14} + \mathrm{i}\sin\frac{54\pi}{14}}{\cos\left(-\frac{12\pi}{14}\right) + \mathrm{i}\left(-\frac{12\pi}{14}\right)}$$
$$= \cos\left(\frac{54\pi}{14} - -\frac{12\pi}{14}\right) + \mathrm{i}\sin\left(\frac{54\pi}{14} - -\frac{12\pi}{14}\right)$$
$$= \cos\frac{33\pi}{7} + \mathrm{i}\sin\frac{33\pi}{7}$$
$$= \cos\frac{5\pi}{7} + \mathrm{i}\sin\frac{5\pi}{7}$$

15)

Let $z = a + b\text{i} = r^r(\cos\theta + \text{i}\sin\theta)$

So that $(a + b\text{i})^n = r^n(\cos\theta + \text{i}\sin\theta)^n = r^n(\cos n\theta + \text{i}\sin n\theta)$

And $(a - b\text{i})^n = r^n(\cos\theta - \text{i}\sin\theta)^n = r^n(\cos(-\theta) + \text{i}\sin(-\theta))^n$

$= r^n(\cos(-n\theta) + \text{i}\sin(-n\theta))$

$\therefore (a + b\text{i})^n + (a - b\text{i})^n = r^n(\cos n\theta + \text{i}\sin n\theta + \cos(-n\theta) + \text{i}\sin(-n\theta))$

$= r^n(\cos n\theta + \text{i}\sin n\theta + \cos n\theta - \text{i}\sin n\theta)$

$= 2r^n\cos n\theta$ which is real.

16)

$\sin\pi = 0$ so since $\left(\cos\frac{\pi}{6} + \text{i}\sin\frac{\pi}{6}\right)^6 = \cos\pi + \text{i}\sin\pi$

$n = 6$

17) a)

$\left(\cos\frac{4\pi}{5} + \text{i}\sin\frac{4\pi}{5}\right)^n = \cos\frac{4n\pi}{5} + \text{i}\sin\frac{4n\pi}{5}$

$\sin\frac{4n\pi}{5} = 0$ so $\frac{4n\pi}{5} = k\pi$ for any integer k

$n = \frac{5}{4}k$ so choose $k = 4$

$\therefore n = 5$

17) b)

$z = 3^5\cos 4\pi = 243$

18) a)

Let $z_1 = \sqrt{3} - \text{i}$ so that $|z_1| = \sqrt{3 + (-1)^2} = 2$ and $\arg z_1 = -\arctan\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$

$z_1 = 2\left(\cos\left(-\frac{\pi}{6}\right) + \text{i}\sin\left(-\frac{\pi}{6}\right)\right)$

Let $z_2 = \sqrt{3} + \text{i}$ so that $|z_2| = \sqrt{3 + 1^2} = 2$ and $\arg z_2 = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$z_2 = 2\left(\cos\frac{\pi}{6} + \text{i}\sin\frac{\pi}{6}\right)$

$z = \frac{z_1}{z_2} = \frac{2\left(\cos\left(-\frac{\pi}{6}\right) + \text{i}\sin\left(-\frac{\pi}{6}\right)\right)}{2\left(\cos\frac{\pi}{6} + \text{i}\sin\frac{\pi}{6}\right)} = \cos\left(-\frac{\pi}{3}\right) + \text{i}\sin\left(-\frac{\pi}{3}\right)$

18) b)

$z^n = \left[\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right]^n$

$\sin\left(-\frac{n\pi}{3}\right) = 0$ so $-\frac{n\pi}{3} = k\pi$ for any integer k

$n = -3k$ so choose $k = -1$

$\therefore n = 3$

19) a)

$|z| = \sqrt{2^2 + 2^2} = 2\sqrt{2}$ and $\arg z = \arctan\left(\frac{2}{2}\right) = \frac{\pi}{4}$

$z = 2\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

19) b)

$z^n = \left[2\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\right]^n$

$\cos\frac{n\pi}{4} = 0$ so $\frac{n\pi}{4} = k\pi + \frac{\pi}{2}$ for any integer k

$n = 4k + 2$ so choose $k = 0$

$\therefore n = 2$

20) a)

$$\frac{\cos\frac{m\pi}{6} + \mathrm{i}\sin\frac{m\pi}{6}}{\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}} = \cos\left(\frac{m\pi}{6} - \frac{n\pi}{4}\right) + \mathrm{i}\sin\left(\frac{m\pi}{6} - \frac{n\pi}{4}\right)$$

$\cos\left(\frac{m\pi}{6} - \frac{n\pi}{4}\right) = 0$ so $\frac{m\pi}{6} - \frac{n\pi}{4} = k\pi + \frac{\pi}{2}$ for any integer k

$\frac{m}{6} - \frac{n}{4} = k + \frac{1}{2} \Rightarrow 2m - 3n = 12k + 6$ so choose $k = 0$

$2m - 3n = 6$

When $n = 2$, $m = 6$

$\therefore n = 2$

20) b)

For z to be real, $\sin\theta = 0$ so $\theta = k\pi$ for some integer k

$m = \frac{3n}{2} + 6k$

n must be even, so that $n = 2$. With $k = 0$ then $m = 3$

So $z = 1$

21) a)

$\alpha\beta\gamma = 4$, $\alpha\beta + \alpha\gamma + \beta\gamma = q$, $\alpha + \beta + \gamma = p$

Since $1 + \mathrm{i}$ is a root, then $1 - \mathrm{i}$ is also a root.

So $(1 + \mathrm{i})(1 - \mathrm{i})\gamma = 4 \Rightarrow 2\gamma = 4$

$p = 1 + \mathrm{i} + 1 - \mathrm{i} + 2 = 4$ and $q = (1 + \mathrm{i})(1 - \mathrm{i}) + 2(1 + \mathrm{i}) + 2(1 - \mathrm{i}) = 6$

21) b)

The three roots are $1 + \mathrm{i}$, $1 - \mathrm{i}$ and 2

Let $z = 1 + \mathrm{i}$ so that $|z| = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\arg z = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$

$z = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)$

So $(1 + \mathrm{i})^n = \left[\sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right)\right]^n = \left(\sqrt{2}\right)^n\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right)$

Let $z = 1 - \mathrm{i}$ so that $|z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$ and $\arg z = -\arctan\left(\frac{1}{1}\right) = -\frac{\pi}{4}$

$z = \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

So $(1 - \mathrm{i})^n = \left[\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)\right]^n = \left(\sqrt{2}\right)^n\left(\cos\left(-\frac{n\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{n\pi}{4}\right)\right)$

So

$$(1 + \mathrm{i})^n + (1 - \mathrm{i})^n + 2^n = 2^{\frac{n}{2}}\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right) + 2^{\frac{n}{2}}\left(\cos\left(-\frac{n\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{n\pi}{4}\right)\right) + 2^n$$

$$= 2^{\frac{n}{2}}\left(\cos\frac{n\pi}{4} + \mathrm{i}\sin\frac{n\pi}{4}\right) + 2^{\frac{n}{2}}\left(\cos\frac{n\pi}{4} - \mathrm{i}\sin\frac{n\pi}{4}\right) + 2^n$$

$$= 2 \times 2^{\frac{n}{2}}\cos\frac{n\pi}{4} + 2^n$$

$$= 2^{\frac{n}{2}+1}\cos\frac{n\pi}{4} + 2^n$$

22) a)

$\alpha\beta\gamma = q$, $\alpha\beta + \alpha\gamma + \beta\gamma = 12$, $\alpha + \beta + \gamma = p$

Since $1 + \sqrt{3}\mathrm{i}$ is a root, then $1 - \sqrt{3}\mathrm{i}$ is also a root.

So $\left(1 + \sqrt{3}\mathrm{i}\right)\left(1 - \sqrt{3}\mathrm{i}\right) + \left(1 + \sqrt{3}\mathrm{i}\right)\gamma + \left(1 - \sqrt{3}\mathrm{i}\right)\gamma = 13 \Rightarrow 4 + 2\gamma = 12 \Rightarrow \gamma = 4$

$p = 1 + \sqrt{3}\mathrm{i} + 1 - \sqrt{3}\mathrm{i} + 4 = 6$ and $q = 4\left(1 + \sqrt{3}\mathrm{i}\right)\left(1 - \sqrt{3}\mathrm{i}\right) = 16$

22) b)

The three roots are $1+\sqrt{3}\mathrm{i}$, $1-\sqrt{3}\mathrm{i}$ and 4

Let $z=1+\sqrt{3}\mathrm{i}$ so that $|z|=\sqrt{1^2+\sqrt{3}^2}=2$ and $\arg z=\arctan\left(\frac{\sqrt{3}}{1}\right)=\frac{\pi}{3}$

$z=2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$

So $1+\sqrt{3}\mathrm{i}=\left[2\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\right]^n=2^n\left(\cos\frac{n\pi}{3}+\mathrm{i}\sin\frac{n\pi}{3}\right)$

Let $z=1-\sqrt{3}\mathrm{i}$ so that $|z|=\sqrt{1^2+\sqrt{3}^2}=2$ and $\arg z=-\arctan\left(\frac{\sqrt{3}}{1}\right)=-\frac{\pi}{3}$

$z=2\left(\cos\left(-\frac{\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)$

So $\left(1-\sqrt{3}\mathrm{i}\right)^n=\left[2\left(\cos\left(-\frac{\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right)\right]^n=2^n\left(\cos\left(-\frac{n\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{n\pi}{3}\right)\right)$

So

$\left(1+\sqrt{3}\mathrm{i}\right)^n+\left(1-\sqrt{3}\mathrm{i}\right)^n+4^n$

$=2^n\left(\cos\frac{n\pi}{3}+\mathrm{i}\sin\frac{n\pi}{3}\right)+2^n\left(\cos\left(-\frac{n\pi}{3}\right)+\mathrm{i}\sin\left(-\frac{n\pi}{3}\right)\right)+4^n$

$=2^n\left(\cos\frac{n\pi}{3}+\mathrm{i}\sin\frac{n\pi}{3}\right)+2^n\left(\cos\frac{n\pi}{3}-\mathrm{i}\sin\frac{n\pi}{3}\right)+4^n$

$=2\times2^n\cos\frac{n\pi}{3}+4^n$

$=2^{n+1}\cos\frac{n\pi}{3}+2^{2n}$

23) a)

$(\cosh z+\sinh z)^n=\left[\frac{1}{2}(\mathrm{e}^z+\mathrm{e}^{-z})+\frac{1}{2}(\mathrm{e}^z-\mathrm{e}^{-z})\right]^n=\left[\frac{1}{2}\mathrm{e}^z+\frac{1}{2}\mathrm{e}^z\right]^n=\mathrm{e}^{nz}$

But since $\cosh nz=\frac{1}{2}(\mathrm{e}^{nz}+\mathrm{e}^{-\mathrm{n}z})$ and $\sinh z=\frac{1}{2}(\mathrm{e}^{nz}-\mathrm{e}^{-\mathrm{n}z})$

Then $\cosh nz+\sinh nz=\frac{1}{2}(\mathrm{e}^{nz}+\mathrm{e}^{-\mathrm{n}z})+\frac{1}{2}(\mathrm{e}^{nz}-\mathrm{e}^{-\mathrm{n}z})=\mathrm{e}^{nz}$

$\therefore(\cosh z+\sinh z)^n=\cosh nz+\sinh nz$ as req.

23) b)

$\cosh(\ln 2) = \frac{1}{2}\left(2 + \frac{1}{2}\right) = \frac{5}{4}$

$\sinh(\ln 2) = \frac{1}{2}\left(2 - \frac{1}{2}\right) = \frac{3}{4}$

$\cosh(3\ln 2) + \sinh(3\ln 2) = (\cosh(\ln 2) + \sinh(\ln 2))^3 = \left(\frac{5}{4} + \frac{3}{4}\right)^3 = 8$

Alternatively:

$\cosh(3\ln 2) + \sinh(3\ln 2) = (\cosh(\ln 2) + \sinh(\ln 2))^3 = \mathrm{e}^{3\ln 2} = 8$

24) a)

$z^2 = \left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)^2 = \cos\pi + \mathrm{i}\sin\pi$

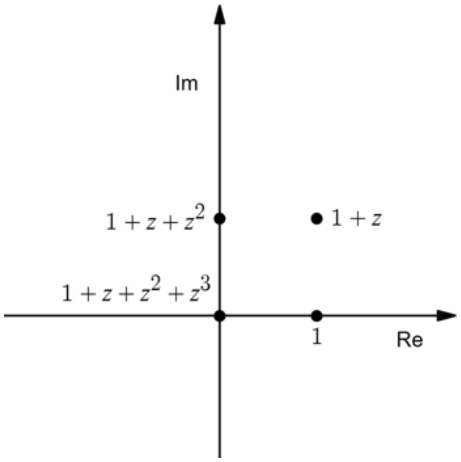
$z^3 = \left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)^3 = \cos\frac{3\pi}{2} + \mathrm{i}\sin\frac{3\pi}{2} = \cos\left(-\frac{\pi}{2}\right) + \mathrm{i}\sin\left(-\frac{\pi}{2}\right)$

24) b)

$1 + z = 1 + \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2} = 1 + \mathrm{i}$

$1 + z + z^2 = 1 + \mathrm{i} + \cos\pi + \mathrm{i}\sin\pi = \mathrm{i}$

$1 + z + z^2 + z^3 = \mathrm{i} + \cos\left(-\frac{\pi}{2}\right) + \mathrm{i}\sin\left(-\frac{\pi}{2}\right) = 0$



24) c) i)

$\arg(1 + z) = \frac{\pi}{4}$

24) c) ii)

$\arg(1 + z + z^2) = \frac{\pi}{2}$